

A NOVEL RLHF APPROACH

DIATool-DPO MULTI-TURN DPO FOR Tool-AUGMENTED LLMs

Enhancing TA-LLM dialogue via automatic preference optimization.

For ML researchers & NLP engineers building production tool agents



SFT Alone Fails in Complex Flow

TA-LLMs trained only with Supervised Fine-Tuning (SFT) struggle with edge cases, leading to **hallucinated tool calls**.

Three Core Failure Modes

- Failing to ask follow-up questions for missing arguments
- Making hallucinated tool invocations to fill gaps
- Failing to reject inappropriate/out-of-scope requests

The Solution: Direct Preference

SFT provides the format, but **preference optimization** teaches the conversational skills of *when* to execute, *when* to ask, and *when* to reject.



5-State Markov Decision Process

Modeling TA-LLM interactions mathematically as 5 internal (unobserved) states.

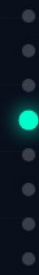
- **1. Initial State:** No history. Out-of-scope requests reject here.
- **2. Tool Selected (w/o Complete Slots):** Missing params. Slot-filling QA occurs here.
- **3. Tool Selected (w/ Complete Slots):** Ready for execution.
- **4. Wait for Tool Response:** Call executed, awaiting results.
- **5. Complete:** Results received, generating final answer.



Three Query Types Define Trajectories

Classifying user requests to map state transitions perfectly.

QUERY TYPE	INFO STATUS	TRAJECTORY	EXAMPLE INTERACTION
Type 1	Sufficient info	1 → 3 → 4 → 5	"Book a flight to NYC tomorrow." (Executes immediately)
Type 2	Incomplete info	1 → (2*N) → 3 → 4 → 5	"Book a flight." → "Where to?" → "NYC" (Slot filling)
Type 3	Out-of-scope	1 → 1	"Write me a poem." (Rejects tool usage)



Automatic Paired Trajectory Construction

No human labels required. The MDP pairs correct (chosen) and incorrect (rejected) dialogue flows automatically.

TYPE	CHOSEN TRAJ.	REJECTED TRAJ.	LEARNING LESSON
Type 1	1 → 3 → 4 → 5	1 → 2 → 3 → 4 → 5	Prevent redundant slot-filling
Type 2	1 → 2 → 3 → 4 → 5	1 → 3 → 4 → 5	Prevent slot hallucination
Type 1/2	Valid Call	1 → 1	Tool call accept
Type 3	1 → 1	1 → 3 → 4	Tool call reject

Dataset includes over 10,000 paired combinations across Easy/Hard difficulties.



SFT + DiaTool-DPO Achieves Best Macro Average

MODEL	METHOD	CALL	COMPLETION	SLOT	RELEVANCE	MICRO AVG	MACRO AVG
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904



LLaMA-3-8B: Dramatic Improvements

Adding DiaTool-DPO to SFT yields massive gains in difficult dialog tasks.



Key Takeaways for Production TA-LLMs

- > **MDP Formulation is Key:** Defining 5 internal states enables systematic, automated preference pair construction without costly human labeling.
- > **DPO Complements SFT:** DPO alone fails, but SFT + DPO excels. Supervised learning provides structure; DPO refines the conversational decision boundaries.
- > **Language Agnostic:** Demonstrated success primarily in Korean but method logic applies universally.
- > **Unlocks Small Models:** Achieves enterprise-grade conversational tool usage (near GPT-4o level) on open-source 8B models.

THANK YOU

